

SR NO	THERMAL & ACOUSTIC INSULATION	UOM	QTY
a	Supply and installation of acoustic lining with in SUPPLY 1st 18meter length ,RETURN air/CTA OR CTS sheet metal duct with class "1" type open cell nitrile insulation 25mm thk MINIMUM density140- 160 kg / m ³ as per ASTM-D-1667 as per the approved shop drawings / specifications. LOW VOC adhesive to be used as per LEED 4.1, VOC of non flammable adhesive shud be less than 50g/l ,(EPD certificate is required for LEED) No additional payment will be made for insulating flanges and coupling etc.	Sqm	1242
b	Supply and installation of acoustic lining with in VAV 1st 1meter length sheet metal duct with class "1" type open cell nitrile insulation 15mm thk MINIMUM density140- 160 kg / m ³ as per ASTM-D-1667 as per the approved shop drawings / specifications. LOW VOC adhesive to be used as per LEED 4.1, VOC of non flammable adhesive shud be less than 50g/l ,(EPD certificate is required for LEED) No additional payment will be made for insulating flanges and coupling etc.	Sqm	379
c	SITC of Thermal Insulation of duct using Close Cell Nitrile rubber 9 mmthickness With density between 40-60 kg / m ³ (External or internal), for rectangle duct ,Joints should be sealed with 50 mm wide 3 mm thick self adhesive tape- LOW VOC adhesive to be used as per LEED 4.1 (similar to insulation),Material shall be FM GLOBAL approved/Greenguard GOLD certified , with tender (Class "0" fire performance according to BS 476 part 6 and part 7 that complies to spread of flame and fire propagation).(VOC of non flammable adhesive to be less than 50g/l)	Sqm	3356

e	SITC of Thermal Insulation for Kitchen TFA of duct using Close Cell Nitrile rubber 13 mm thickness With density between 40-60 kg / m³(External or internal), for rectangle duct ,Joints should be sealed with 50 mm wide 3 mm thick self adhesive tape- LOW VOC adhesive to be used as per LEED 4.1 (similar to insulation),Material shall be FM GLOBAL approved/Greenguard GOLD certified, with tender (Class "0" fire performance according to BS 476 part 6 and part 7 that complies to spread of flame and fire propagation).(VOC of non flamable adhesive to be less than 50g/l)	Sqm	45
	SITC of Acoustic Lining of FAN/AHU Room(wall and slab) using Rigid board No/Free formaldehyde(Non itching) fibre glass with 50 mm thick 48 kg / m³ fitted in 0.6 mm thick G.I. Sheet channel frame make and covered with RP tissue paper & 22 g GI POWDER COATED perforated sheet Minium opening of perforated sheet shall be 60% ,as detailed in the specifications complete with GI beading (vertical&horizontal)on the 600 by 600 grid.(Greengard certified material need to used as per LEED)	Sqm	1021
	SITC of Mild Steel Chilled Water Piping of class 'C', IS 1239 & 3589with Nitrile rubber Insulation ,		
	Isulation shall be plain type nitrile rubber as per below thick specified.		
	Up to Dia 80 mm Dia 32 mm thick .		

	100 & ABOVE 38mm (19MM)two layers sheet to be usedInsulation installation process / methodology and finish shall be as per tender specs , Close Cell Nitrile rubber insulation glass fabric (thickness of cover 6-7 mils minimum or 200 gm/sqm+-10%) Class 'O FM GLOBAL APPROVED(COLOUR WILL BE RED, BLUE).with Pipe to cut to required length, and installed with welded joints, including all necessary fittings such as elbows, tees, bends, reducers, flanges, U clamps with gasket, supports duly insulated as indicated in the insulation specification section. Necessary pipe supports to be provided as per the spec. refer tender specs for details.		
	The bare piping shall be painted with Zinchromate primer (yellow primer)- LOW VOC as per LEED 4.1 , before installation of piping & Insulation		
a	25 mm - 32 mm thick	Rmt	310
b	32 mm - 32 mm thick	Rmt	330
c	40 mm - 32 mm thick	Rmt	10
d	50 mm - 32 mm thick	Rmt	108
	Supply & Installation of Hard copper piping (Near idu termination soft copper allowed) duly insulated with nitrile insulation laminated with glass fabric (thickness of cover 6-7 mils minimum or 200 gm/sqm+-10%)RED/ YELLOW TUFF COATING of 19mm thickness for Gas Pipe and insulated with nitrile insulation BLUE TUFF COATING(IC CLAD OR ARMA CHECK) 13mm thickness for Liquid Pipe / Including supports Clamps etc and incl bends, couplings and necessary fittings, for R-410A; Site Pressure tested to 550 psig		
	Dia - 6.4 mm	Rmt	98
	19 mm Class O Nitrile rubber GC sheet	Sqm	14
	Dia - 9.5 mm	Rmt	289
	Dia - 12.7 mm	Rmt	536
	Dia - 15.9 mm	Rmt	364
	Dia - 19.1 mm	Rmt	69

	Dia - 22.2 mm	Rmt	56
	Dia - 25.4 mm	Rmt	5
	Dia - 28.6 mm	Rmt	531
	Dia - 31.8 mm	Rmt	2
	Dia - 34.9 mm	Rmt	21
	Dia - 38.1 mm	Rmt	2
	SITC of drain piping of CPVC SDR13.5 with 6 mm thick nitrile foam insulation laminated with GC glass fabric (thickness of cover 6-7 mils minimum or 200 gm/sqm+-10%) (white)& all fittings, supports and clamps up to 50 mm dia. including drilling, wall chipping etc and conceal the wall opening.(OFFICE FLOOR) - <u>9 MM THICKNESS CONSIDERED</u>		
	25 mm dia	Rmt	183
	32 mm dia	Rmt	142
	40 mm dia	Rmt	5
	50 mm dia	Rmt	5